5G-A RedCap: Technology and Applications

Hua Zhang
Research Institute
China Telecom
Beijing, China
zhanghua111@chinatelecom.cn

***Abstract*—5G-A Reduced Capability (RedCap), defined in the 3rd Generation Partnership Project (3GPP) Release 17 standard, is a mid-speed Internet of Things (IoT) technology that fills the technological gap between high-speed 5G eMBB and low-speed NB-IoT. By streamlining functions, it reduces terminal complexity by 60% - 70% while meeting the demands of mid-speed IoT scenarios. Its core design objectives include reducing power consumption, optimizing costs, and enhancing coverage. This paper systematically reviews the standardization process and recent advancements of RedCap, analyzes typical business scenarios, details its fundamental technical characteristics, performance, and enhancement solutions, and presents experimental results. Finally, it outlines the key challenges and future evolution paths for 5G RedCap technology.**

Keywords—RedCap, IoT, BWP, SUL

I. INTRODUCTION

In June 2020, 3GPP officially initiated the work on standardizing 5G RedCap technology. The primary research focused on reducing terminal complexity and associated cost assessments, evaluating impacts on network coverage, and conducting energy consumption analysis. By June 2022, the first version of RedCap was finalized as part of the 3GPP Release 17 (R17) standard. This release addressed multiple technical aspects, including reduction of terminal complexity, residence and access control, mobility and terminal identification, Bandwidth part (BWP) configuration, and power consumption control. In June 2024, an enhanced version of RedCap was completed with 3GPP R18, further expanding its scope to include energy saving and efficiency improvements, enhanced Extended Discontinuous Reception (eDRX), and complexity and cost reduction in Frequency Range 1 (FR1).

Existing Internet of Things (IoT) technologies such as Narrow Band Internet of Things (NB-IoT) and LTE-Machine-to-Machine (LTE-M) offer multiple advantages: they are based on open standards, provide global coverage, are vendor-independent, operate on licensed spectrum, and support scalable deployment. These technologies not only extend battery life but also significantly enhance network coverage. In comparison, RedCap supports higher data rates than Low Power Wide Area (LPWA) solutions like NB-IoT and LTE-M, while maintaining considerably lower cost and power consumption than traditional NR implementations. This enables an effective balance between network performance and terminal cost.

The introduction of RedCap technology bridges a critical gap in 5G applications for mid- to high-speed scenarios. It establishes a comprehensive mobile IoT technical architecture capable of meeting diverse service requirements across low-, medium-, and high-speed use cases, while also facilitating coordinated development between 4G and 5G networks.

As a lightweight 5G technology, RedCap achieves cost and power consumption reductions by appropriately scaling down certain performance metrics, while retaining key 5G advantages such as high data rates and low latency. Its specific technical characteristics define its applicable scenarios. Driven by practical service requirements, RedCap seeks an optimal trade-off between cost and performance. By reducing terminal complexity to lower costs, it promotes the large-scale adoption of 5G across a wider range of fields. Moreover, the tailored terminal performance must still satisfy diverse industrial requirements. For example, industrial applications such as power systems impose strict demands for high reliability and low latency; video surveillance services require high network capacity and multi-user concurrency support; industrial campuses emphasize data privacy and security, such as ensuring data remains on-premises; and wearable smart devices impose stringent requirements for compact form factors and low power consumption.

Reference [1] evaluates the coverage of NR RedCap across various channels and scenarios. Results indicate that the required coverage recovery is minimal (under 1 dB) and can be offset by adjusting data rate targets, with no channels identified as severely coverage-limiting.

Reference [2] summarizes 3GPP R18's positioning solutions for RedCap devices, evaluates the achievable positioning accuracy and presents a comparative performance analysis through simulations.

Reference [3] analyzes the performance of eigenmode beamforming for 5G RedCap devices utilizing single-layer transmission, and the effect of the number of UE antennas on the achievable rate is investigated.

Reference [4] provides a comprehensive overview of 3GPP R17 RedCap while describing newly introduced features, cost reduction and power saving gains, and performance and coexistence impacts.

Building on the studies of fundamental RedCap characteristics and simulation results, this paper focuses on enhanced technical solutions and systematically presents performance and functional test results of 5G RedCap, including its coverage performance, latency performance, and service compatibility across multiple frequency bands. Finally, it analyzes the key challenges and future development paths facing 5G RedCap technology.

II. Technical Characteristics and Performance

The standardized design of 3GPP RedCap terminals aims to lower the cost of 5G devices while still meeting user communication requirements in various 5G application scenarios. It also strives to maintain a high degree of compatibility with existing 5G systems. To achieve these objectives, RedCap has introduced optimizations primarily in hardware simplification, access and control protocols, and terminal energy efficiency [5].

A. Reduced terminal complexity

In terms of radio capability, RedCap supports a reduced maximum bandwidth—from 100 MHz to 20 MHz in FR1 and from 400 MHz to 100 MHz in FR2. This bandwidth reduction relaxes the baseband processing requirements, thereby lowering device cost. RedCap offers a downlink peak data rate of up to 150 Mbps and an uplink peak rate of 50 Mbps, which meets the transmission rate requirements of mid-tier applications while avoiding resource over-provisioning. Compared with enhanced Mobile Broadband (eMBB), which demands ultra-high bandwidth, RedCap achieves a better balance between cost and performance through more reasonable bandwidth allocation. At the same time, it significantly enhances data transmission capabilities compared to LPWA technologies, providing robust support for applications such as industrial monitoring and smart logistics that require certain levels of data real-time performance.

B. Residence and Access Control

To enable more flexible network control and prevent potential interference with legacy NR terminals, specific camping control mechanisms have been designed for RedCap devices. These mechanisms allow the network to restrict RedCap terminal access based on its load conditions. RedCap User Equipments (UEs) are permitted to camp only on cells that support them, and the network may impose restrictions on different types of RedCap UEs, such as those with 1Rx or 2Rx configurations. Key access control features include:

The use of IntraFreqReselectionRedCap (IFRI) parameters in system information to indicate whether RedCap UEs are allowed to select or reselect a cell on the same frequency, configurable access acceptance or rejection policies for 1Rx/2Rx RedCap UEs. And the introduction of a Logical Channel ID (LCID) identifier in Msg3/MsgA to indicate and recognize RedCap devices.

C. Efficient Access

For mobility-dependent services such as wearable devices, RedCap provides comprehensive support for cell selection, reselection, and handover procedures. BWP as a distinctive feature of 5G, plays an important role in enabling efficient RedCap terminal access and coexistence with conventional NR terminals. In NR TDD cells, RedCap and standard NR UEs adopt differentiated initial access configurations depending on the bandwidth scenario. For instance:

In a 20 MHz FDD scenario, they share the same initial uplink and downlink BWP.

In a 100 MHz TDD scenario, separate BWPs are configured, with shared PRACH and PUCCH resources within a 20 MHz bandwidth. These BWPs are allocated at both ends of the cell's frequency domain, reducing fragmentation of PUSCH resources for regular UEs.

Such intelligent BWP configuration strategies improve overall network resource utilization while ensuring access efficiency and stability for different types of terminals.

D. Power Consumption Optimization

For power-sensitive devices like wearables, RedCap introduces a range of innovative low-power features. The Radio Resource Management (RRM) measurement relaxation mechanism allows the network to dynamically adjust measurement cycles and parameters according to UE mobility and service needs, thereby reducing unnecessary measurement operations and lowering power consumption. The eDRX feature extends the paging monitoring cycle, allowing the terminal to enter a deep sleep state when not listening for paging. Specifically:

In RRC_IDLE state, the eDRX cycle can be extended up to 10485.76 seconds.

In RRC_INACTIVE state, it can be extended up to 10.24 seconds.

These low-power designs significantly extend battery life, reduce the frequency of charging, lower operational costs, and provide technical support for the widespread adoption of power-constrained applications such as wearables and smart sensors.

E. Multi-mode, Multi-band, and Voice Features

Considering the diversity of communication modes and voice service requirements across different industries, RedCap terminals have been specifically optimized in terms of multi-mode multi-band operation and voice capabilities.

For private network deployments, single-mode terminals can be used to fulfill specific frequency band and service requirements, reducing device cost and complexity.

For services requiring continuous connectivity, 4G/5G multi-mode operation ensures seamless handover across different network environments, enhancing service reliability.

For voice-enabled applications, terminals support Voice over New Radio (VoNR), and several wearable devices already incorporate this functionality to meet basic voice communication needs.

This support for multi-mode multi-band and voice services enables RedCap to be widely deployed across various fields, including the Industrial Internet of Things, intelligent transportation, and healthcare.

III. Enhanced Technical Solutions

A. Coverage Enhancement

In FDD deployment scenarios, both eMBB commercial terminals and RedCap terminals typically feature 1Tx uplink transmission, resulting in comparable uplink coverage capabilities. In TDD deployment scenarios, however, eMBB commercial terminals generally support 2Tx uplink, while RedCap terminals are limited to 1Tx. Consequently, RedCap exhibits inferior uplink coverage compared to eMBB, leading to an approximate 75% reduction in edge data rates.

To enhance RedCap uplink coverage, techniques such as increasing the retransmission count for traffic channels, implementing multi-slot joint channel estimation, and utilizing multi-slot transport block transmission can be employed.

B. Capacity Enhancement

Capacity enhancement is achieved through BWP enhancements, enabling RedCap terminals to coexist within existing R15/R16 networks. These include initial BWP enhancements and dedicated BWP enhancements. A single 20MHz BWP resource is often insufficient to meet demands, and fixed BWP configurations cannot adapt to dynamic capacity requirements. Furthermore, the coexistence of RedCap and eMBB terminals can lead to a decline in overall network spectral efficiency. Network capacity can be improved through multi-BWP strategies with adaptive activation/deactivation, and spatial multiplexing between RedCap and eMBB users.

The initial BWP is used by terminals during the initial access procedure. For RedCap terminals, this is categorized into independent initial BWP and shared initial BWP. Since RedCap terminals support a maximum bandwidth of 20MHz, they can share the initial BWP with eMBB terminals if the network-configured initial BWP for 5G eMBB terminals is less than or equal to 20MHz. However, this may cause congestion as both RedCap and eMBB users concentrate within the same narrowband resources. To address this, the standards introduced an independent initial BWP, allowing the network to configure separate initial uplink and downlink BWPs for RedCap terminals. This introduction also involved Non-Cell Defining SSB (NCD-SSB), as shown in Fig. 1, enabling RedCap UEs to access initial BWPs that do not contain the Cell Defining SSB (CD-SSB). Generally, configuring an independent initial BWP for RedCap UEs is more effective in TDD 5G networks with 100MHz system bandwidth, whereas a shared initial BWP suffices for FDD 5G networks with system bandwidths of 20MHz or slightly larger.

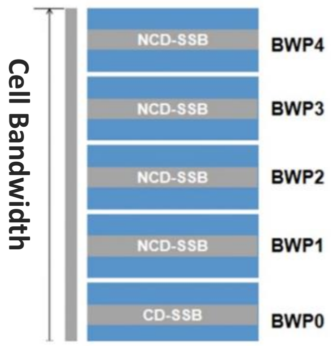

Fig. 1 5G RedCap BWP Configuration

After transitioning to the connected state, RedCap terminals operate on the dedicated BWP configured via RRC. The introduction of NCD-SSB provides significant flexibility in configuring dedicated BWPs in the connected state. RedCap UEs can operate on dedicated BWPs containing either the CD-SSB or an NCD-SSB, supporting measurements for both serving and neighbor cells based on either type. The network can also configure multiple dedicated BWPs containing NCD-SSBs. These can be activated on-demand when the number of RedCap users or the traffic volume is high, enhancing the capacity for RedCap terminals while ensuring the user experience for NR terminals.

C. RedCap SUL(Supplementary Uplink)

Current 5G networks primarily employ NR TDD deployment, where uplink and downlink share the same spectrum resources through time-division duplexing. Due to the limited availability of uplink time-frequency resources, user uplink experience is often constrained. The 5G SUL (Supplementary Uplink) technology addresses this limitation by enabling uplink data transmission to be time-shared between the NR TDD spectrum and the SUL spectrum, significantly expanding the available uplink time-frequency resources for 5G users.

For RedCap terminals operating on TDD carriers, overall performance is constrained by the scarcity of uplink slots and the higher frequency bands. To mitigate this, an additional uplink component—the SUL link—has been introduced within NR TDD cells. Without reducing the downlink bandwidth, the uplink capability is enhanced through time-shared scheduling between the native NR TDD uplink (also referred to as the NUL, or Normal Uplink) and the supplementary SUL link. Specifically, during uplink slots on the NUL, the UE transmits uplink data via the NUL; during downlink or flexible slots on the NUL, the UE utilizes the SUL for uplink data transmission, as illustrated in Fig. 2. By incorporating SUL, uplink data can be transmitted across all slot types, thereby effectively increasing uplink bandwidth without compromising downlink capacity. This enhancement reliably supports the transmission requirements of various bandwidth-intensive uplink services.

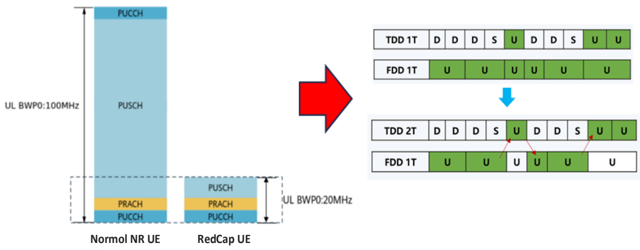

Fig. 2 Scheduling Principle

IV. Field Testing for RedCap Performance Evaluation

To validate the single-site functionality, network performance, and effectiveness of enhanced technical schemes for 5G RedCap, a comprehensive field test was conducted in live network environments. The testing methodology was designed as follows: single-site functionality and mobility tests were performed in a localized area comprising 2–4 base stations, while network-level performance evaluation was carried out in urban or county-scale scenarios with 10–20 contiguous base stations per frequency band across both 4G and 5G systems. The single-site and network test areas were not required to be co-located.

The RedCap terminals deployed in these tests supported non-standard capabilities, including the ability to ignore specific indications in SIB1. These devices operated in LTE/NR dual-mode configuration, with optional support for

SUL functionality. When operating in LTE mode, the terminals complied with Category 4 (CAT4) requirements, as did the reference LTE UE terminals used for comparative analysis.

In the single-site performance comparison, RedCap terminals demonstrated performance comparable to commercially available devices. The detailed experimental data are summarized in Table I.

TABLE I AVERAGE LATENCY

Category		Ping 100 times with 32-byte packets	Ping 100 times with 1500-byte packets
RedCap UE with 5G switched off	Near point	22	29
	Mid- point	24	38
	Far point	34	45
LTE UE	Near point	21	31
	Mid- point	25	36
	Far point	35	50

RedCap UEs delivered comparable performance in 4G networks, achieving uplink and downlink data rates ranging from 88% to 104% of the rates observed for conventional LTE UEs, as depicted in Fig. 3.

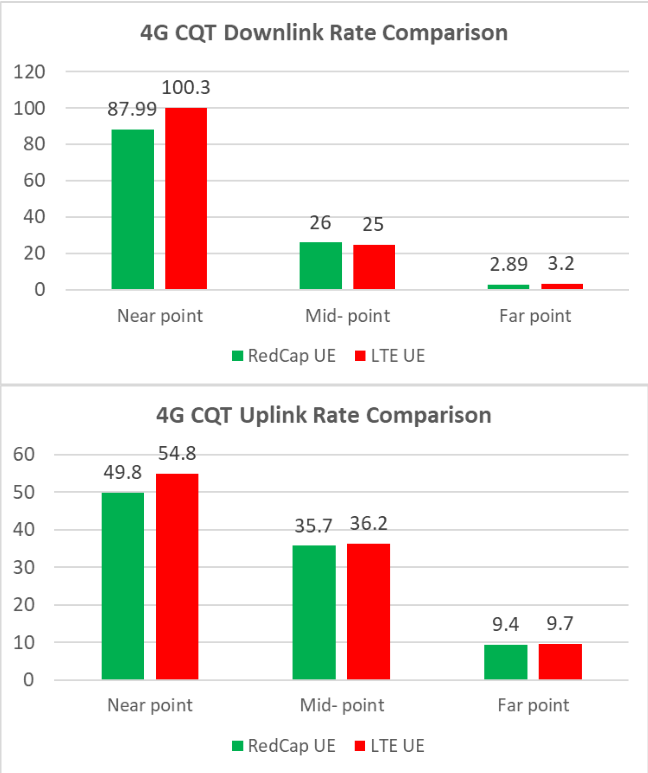

Fig. 3 4G Network CQT Performance

In 5G mode, as shown in Fig. 4, the RedCap UE achieved uplink and downlink data rates of approximately 12% and 10%, respectively, of those attained by the NR UE.

The RedCap UE demonstrates a significant boost in both uplink and downlink throughput under NCD-SSB configuration, while maintaining normal neighbor cell handover capability whether operating in NCD or CD-SSB BWP, as shown in Fig. 5.

Enabling the SUL feature provides RedCap with extended uplink coverage and improved uplink data rates, while maintaining downlink performance. Additionally, it reduces user plane latency by approximately 1–2ms.

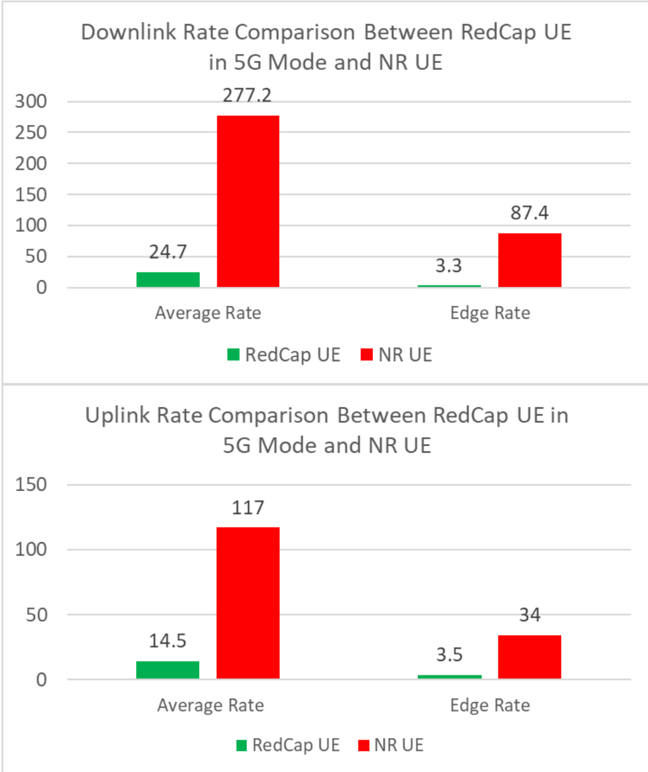

Fig. 4 5G Network DT Performance

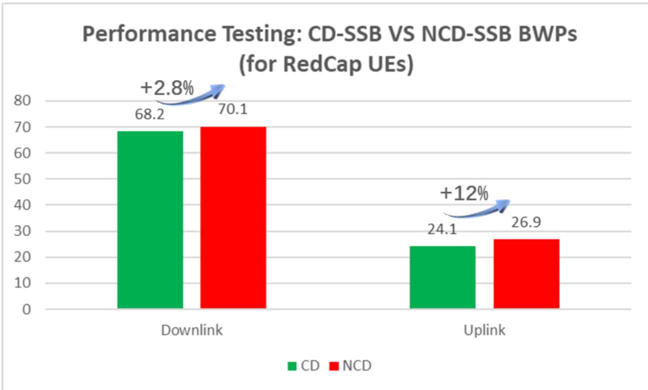

Fig. 5 Performance Evaluation of BWPs with CD-SSB and NCD-SSB

V. CONCLUSION

The 5G RedCap ecosystem, while now commercially viable and gaining initial traction, still faces several critical barriers to mass-scale deployment.

RedCap is primarily targeted at mid- to high-speed IoT scenarios that are price-sensitive, such as wearables and video surveillance. Although terminal costs have been reduced through techniques like bandwidth reduction and fewer transceiver antennas, the current price remains in the range of 200–300 RMB. Future efforts must focus on driving terminal prices below 100 RMB and gradually aligning them with current 4G Cat 4 terminal costs. Consequently, in the early commercial phase, RedCap will face direct competition from cost-effective and technologically mature 4G terminals [6].

The broad adoption of IoT applications and a satisfactory user experience heavily depend on network coverage quality. Despite China having built the world's largest 5G network, its deep coverage still lags behind that of 4G networks, particularly in complex scenarios such as indoor environments, where enhancements are still needed.

The IoT industry involves multi-party collaboration among operators, equipment vendors, chip/module manufacturers, and application providers. Interoperability between RedCap networks and traditional networks, as well as the integration of new RedCap terminals with traditional industries, requires a long-term process of adaptation and optimization to achieve balance and mutual success. Moving forward, it is essential to promote open collaboration across the industry chain and foster a cooperative ecosystem to facilitate the successful large-scale commercialization of 5G RedCap technology.

Acknowledgment

This work was supported by National Science and Technology Major Project-Mobile Information Networks under Grants No. 2024ZD1300300.

References

[1] S. Moloudi et al., "Coverage Evaluation for 5G Reduced Capability New Radio (NR-RedCap)," in IEEE Access, vol. 9, pp. 45055-45067, 2021, doi: 10.1109/ACCESS.2021.3066036.

[2] F. Munier et al., "Positioning of RedCap Devices in 5G Networks," in IEEE Communications Magazine, vol. 62, no. 8, pp. 110-116, August 2024, doi: 10.1109/MCOM.001.2300802.

[3] K. Min and T. Kim, "Performance Analysis on Eigenmode Beamforming for Reduced Capability Device in Massive MIMO Systems," in IEEE Access, vol. 11, pp. 13103-13112, 2023, doi: 10.1109/ACCESS.2023.3242865.

[4] S. N. K. Veedu et al., "Toward Smaller and Lower-Cost 5G Devices with Longer Battery Life: An Overview of 3GPP Release 17 RedCap," in IEEE Communications Standards Magazine, vol. 6, no. 3, pp. 84-90, September 2022, doi: 10.1109/MCOMSTD.0001.2200029.

[5] Shi, Y., Zhu, Z. Y., Sun, H. F., et al. Research on 5G RedCap Lightweight Key Technologies and End-to-Network Practice[J]. Information and Communications Technologies, 2024, 18(1): 44-51,78.

[6] Huang, W. J., Zou, D. B., & Zhang, C. X. (2024). Analysis and Performance Verification of 5G RedCap Technology. *Telecommunications Engineering Techniques and Standardization*, 37(3), 7-12+30. https://doi.org/10.13992/j.cnki.tetas.2024.03.013